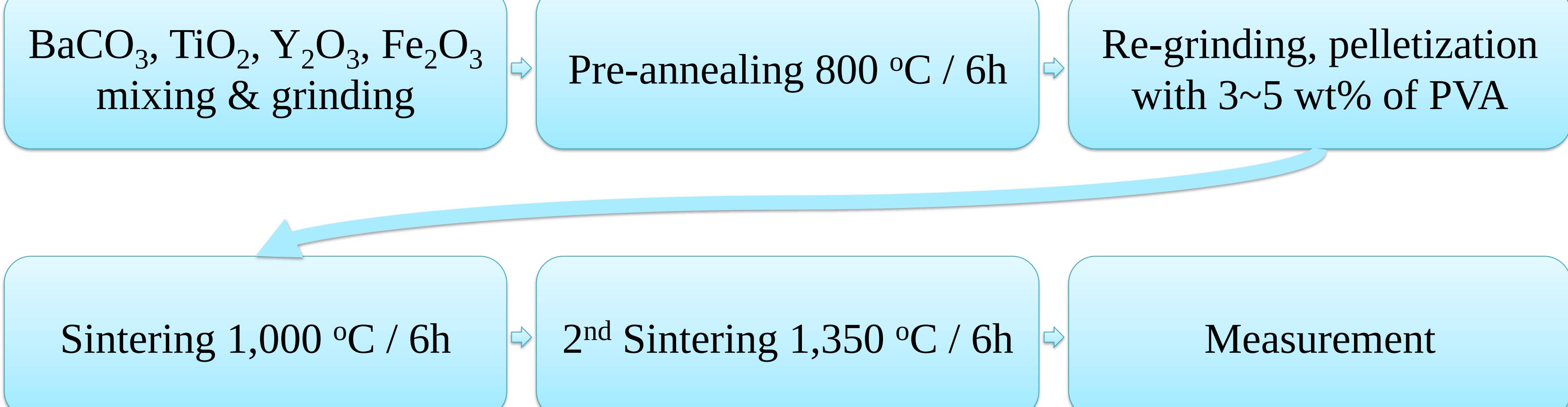


Introduction

❖ Motivation

- ✓ Multiferroics have been known as smart materials showing a co-existence of multiple properties, such as ferroelectricity, ferromagnetism and ferroelasticity, that are applicable for multifunctional electronic devices.
- ✓ Basically, the co-existence of ferroelectricity and ferromagnetism can be carried out through doping magnetic (3d-4f) elements into a ferroelectric host lattice or making a composite of ferroelectric and ferromagnetic materials.
- ✓ In this work, we present the fabrication and consider the possibility of forming solid solutions of Ba_{1-x}Y_xTi_{1-x}Fe_xO₃ (BY_xTF_xO) ceramics.

❖ Experimental Process



Experiments & Results

❖ SEM/EDS

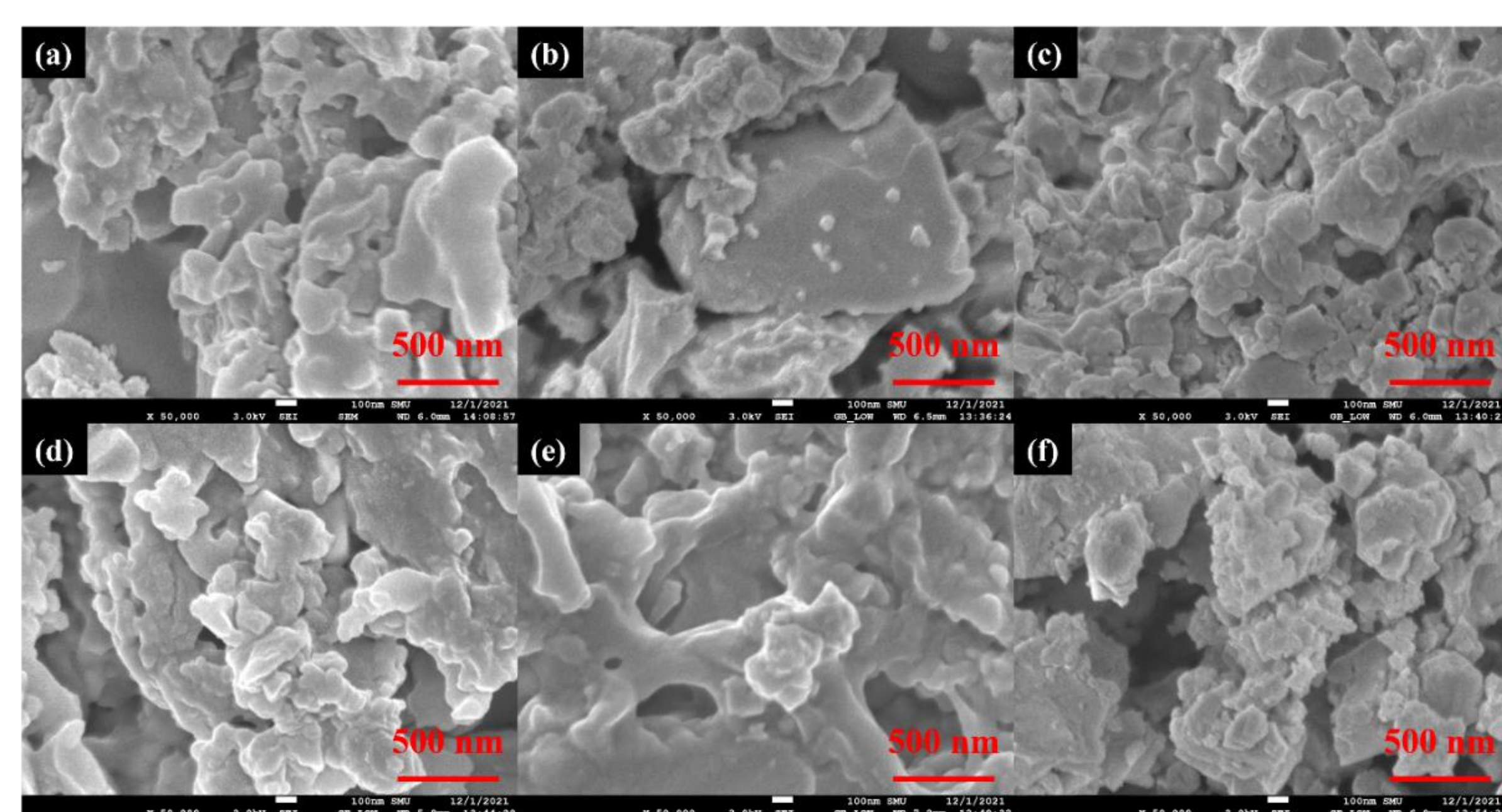


Fig.1. SEM images showing surface morphologies of all as-prepared BY_xTF_xO pellets: (a) $x = 0$, (b) $x = 0.01$, (c) $x = 0.02$, (d) $x = 0.04$, (e) $x = 0.08$, and (f) $x = 0.1$, which were recorded with the same magnification, corresponding a bar scale of 0.5 μm .

- ✓ We have checked the surface morphology of all the fabricated BY_xTF_xO ($x = 0-0.1$) samples.
- ✓ The sample pellets consist of agglomerated particles with irregular shapes.
- ✓ Their morphology remains less changed with respect to doping content x .

❖ Raman Spectra

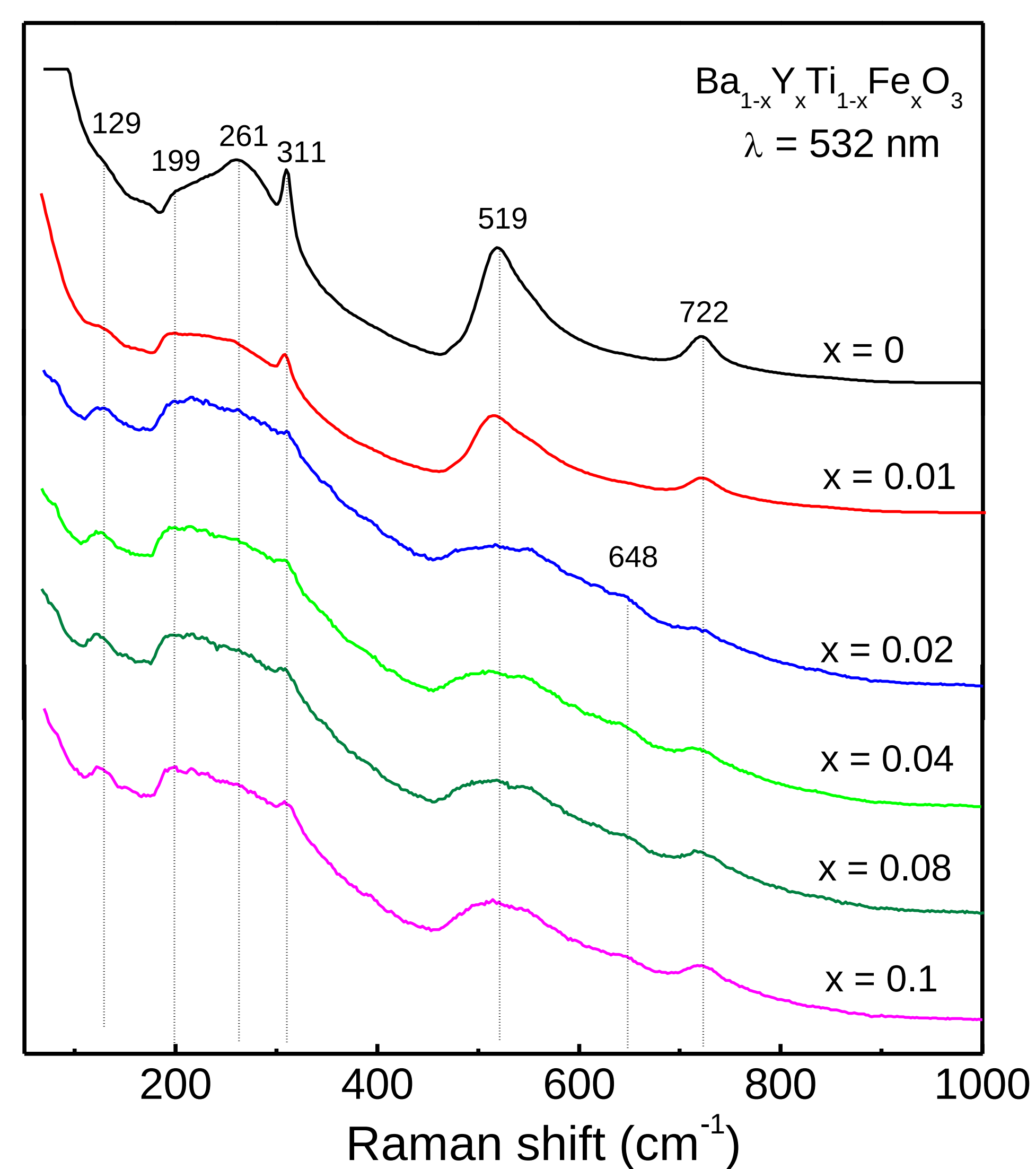


Fig.2. Raman spectra of BY_xTF_xO ceramics excited by $\lambda_{\text{exc}} = 532 \text{ nm}$. The dotted lines show vibrational modes.

- ✓ The 311 cm^{-1} peak can be assigned to the E(LO) and E(TO) mode, and it is related to the tetragonal-cubic phase transition phenomenon.
- ✓ The broad peak near 519, and 722 cm^{-1} are related to the features in tetragonal phase.
- ✓ We can know that the tetragonality of BY_xTF_xO decreases as the x content increases.

❖ X-ray Diffraction (XRD)

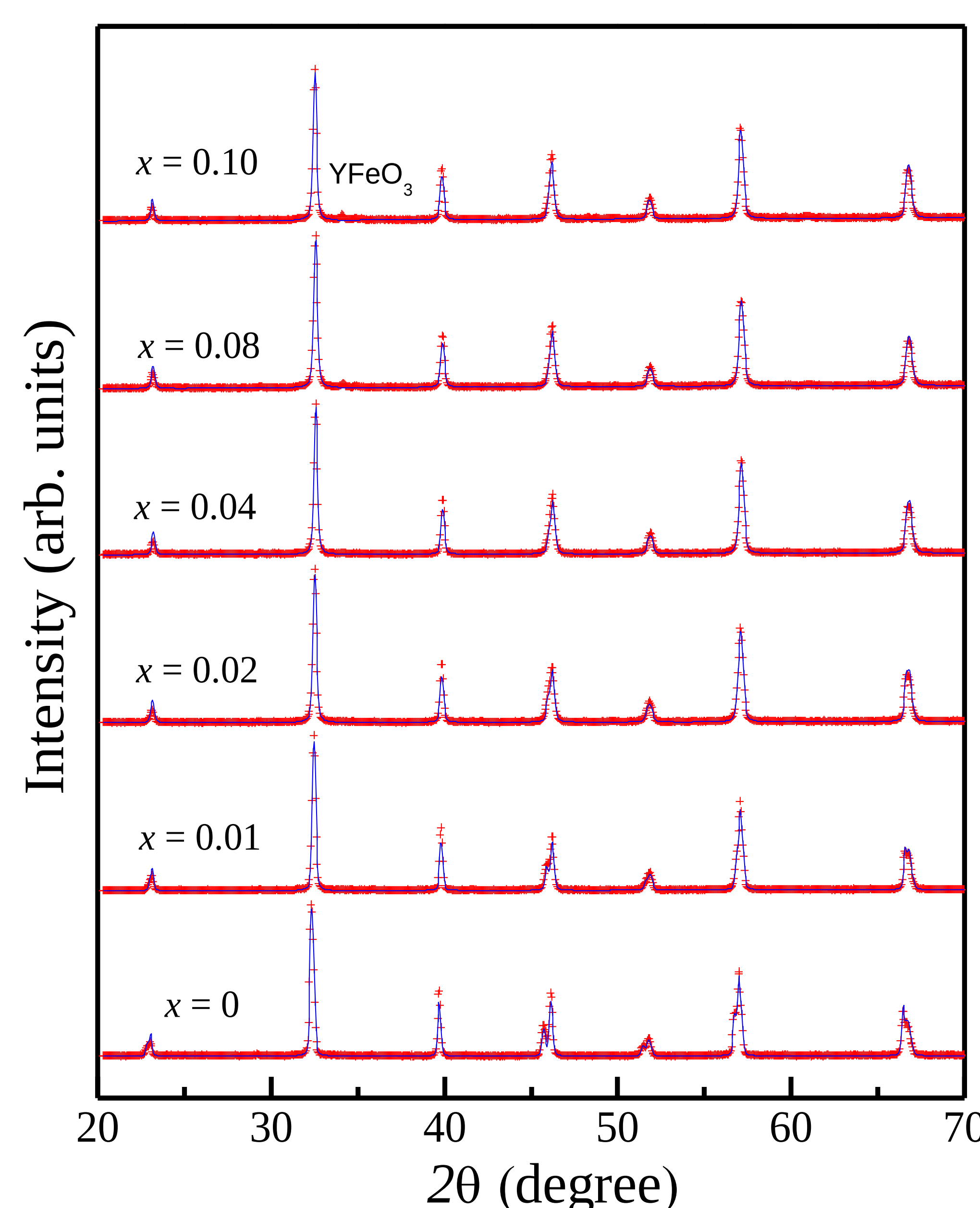


Fig.3. X-ray powder diffraction data of BY_xTF_xO ceramics analyzed as using Rietveld refinement. A small fraction the YFeO₃-related phase starts constituting as $x = 0.08$.

❖ Lattice Parameter

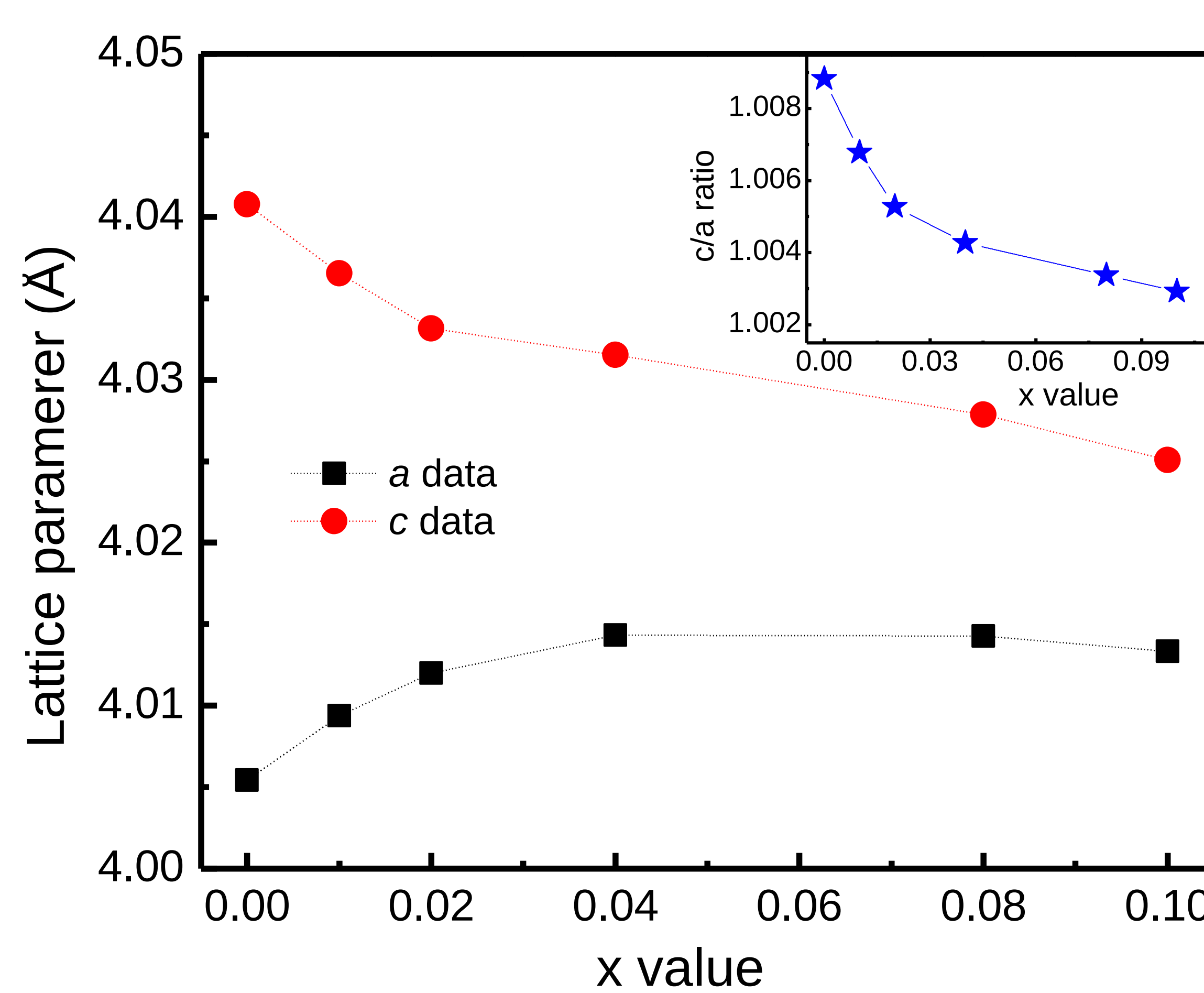


Fig.4. Variations of the lattice constants, a and c , as a function of x in BY_xTF_xO ceramics. The inset graphs the variation of c/a towards the unity of a cubic structure.

- ✓ a gradually increases while c decreases when x increases.
- ✓ Such feature leads to a gradually reduced c/a ratio towards the unity of a cubic structure.

❖ Magnetic Properties

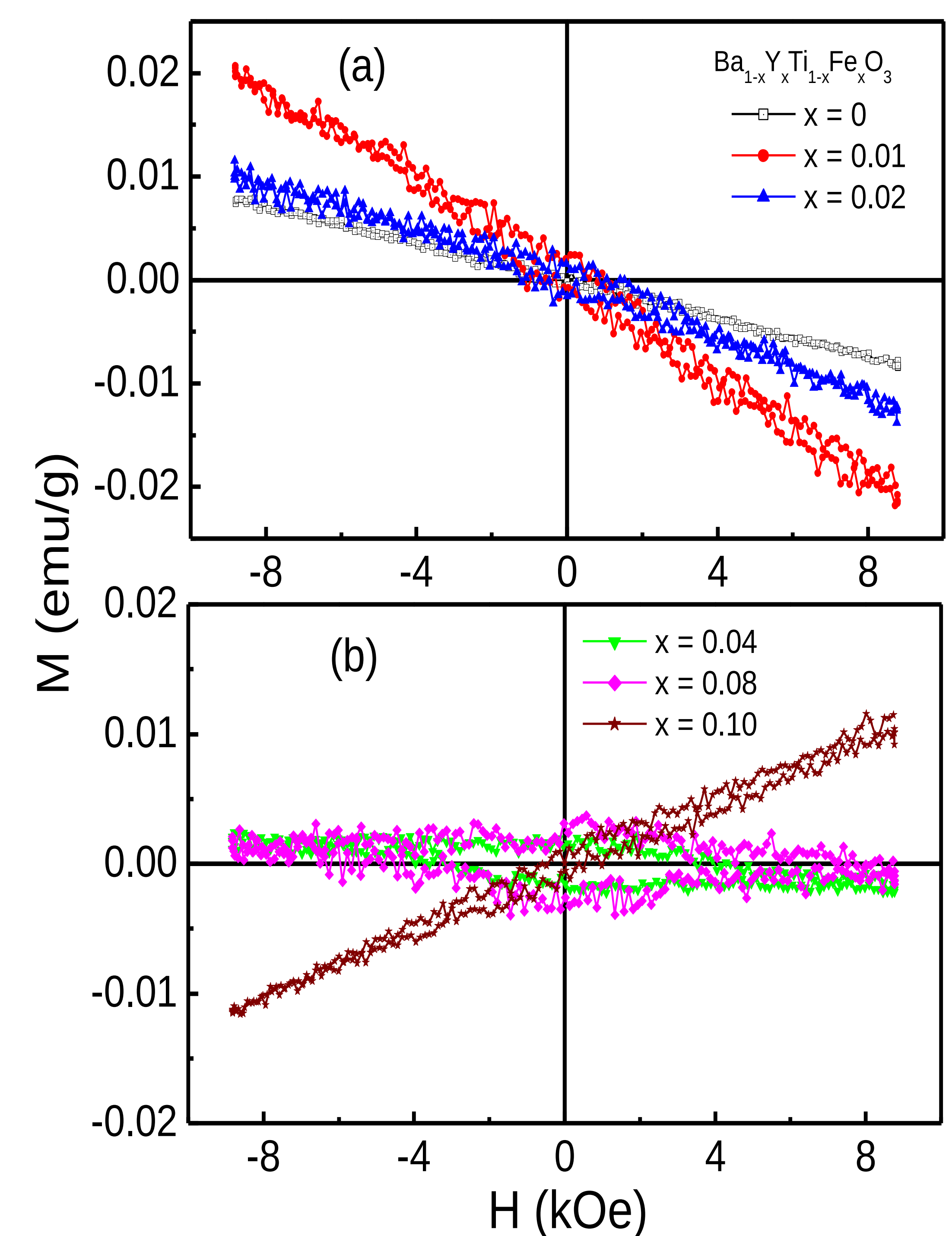


Fig.5. Magnetic hysteresis loops of BY_xTF_xO samples (a) $x = 0-0.02$ (b) $x = 0.04-0.1$

- ✓ Fig.5(a) shows the magnetic hysteresis curve of $x < 0.04$ samples.
- ✓ Fig.5(a) samples show diamagnetism in all magnetic fields.
- ✓ Fig.5(b) shows the magnetic hysteresis curve of $x = 0.04 - 0.1$ samples.
- ✓ Fig.5(b) samples show a coexistence of paramagnetic and weak ferromagnetic behaviors
- ✓ For $x = 0.04$ and $x = 0.08$ samples, they show ferromagnetic properties in the low range of magnetic field.
- ✓ On the other hand, $x = 0.1$ sample shows ferromagnetic characteristic in all magnetic fields.

Conclusions

- ✓ We show a study about the structural characterization, and magnetic properties of Ba_{1-x}Y_xTi_{1-x}Fe_xO₃ (BY_xTF_xO) ceramics.
- ✓ The samples crystallized in a tetragonal structure with the constitution of a YFeO₃-related secondary phase as $x \geq 0.08$
- ✓ With increasing x , the lattice constant a increases while c decreases.
- ✓ Magnetization measurements indicate BY_xTF_xO transiting from the diamagnetic behavior to the paramagnetic one when x increases, with the appearance of weak ferromagnetism that is assigned to super-exchange interactions between Fe³⁺ ions.